

Combinations



1 Evaluate the following expressions:

a $5!$

b $4!6!$

c $\frac{11!}{5!6!}$

d $\frac{19!}{15!4!}$

e $\frac{5!}{4!}$

f $\frac{10!}{5!4!}$

g $\binom{7}{2}$

h 4C_0

i $\binom{5}{5}$

j $\binom{7}{3}$

k $\binom{10}{9}$

l $\frac{{}^{10}C_5}{{}^{10}C_6}$

2 Simplify:

a $n(n-1)!(n+1)$

b $\frac{(n+1)!}{n!}$

c $\frac{n!}{(n-2)!}$

d $\frac{(n+4)!}{(n+6)!}$

3 Find the value of n if:

a $\binom{n}{3} + \binom{n}{2} = 8 \binom{n}{1}$

b The ratio of the number of combinations of $2n + 2$ different objects taken n at a time to the number of combinations of $2n - 2$ different objects taken n at a time is $14 : 1$.

4 Prove the following:

a $\binom{n}{0} = 1$

b $\binom{n}{1} = n$

c $\binom{n}{0} = {}^nC_n$

d $\binom{100}{100} = 1$

e $\binom{100}{100-1} = 100$

f $\binom{100}{100-40} = \binom{100}{40}$

Applications

5 Ben has 3 shirts, each in a different colour: crimson (C), pink (P) and white (W), and 4 ties, each in a different colour: blue (B), grey (G), red (R) and yellow (Y).

a Construct a table to show all the possible combinations Ben could wear.

b How many different combinations are possible?

6 A hotel is supervised by a team of 6 security guards at any given time during the day. If there are 12 security guards available, how many different ways can a team of guards be chosen?

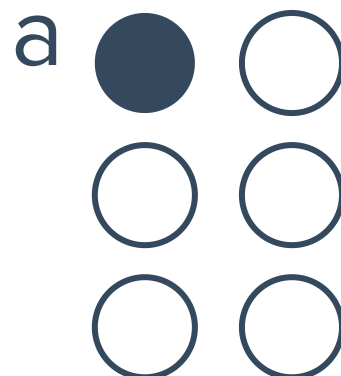
7 A boss wants to select one group of 4 people from his 28 staff. How many different groups



- 8 A group of 28 students visit the tenpin bowling centre and must organise themselves into a group of 8 to use a lane. How many different ways are there for a group of 8 students to be formed?
- 9 A boss wants to select one group of 3 people from his 27 staff. How many different groups are possible?
- 10 A restaurant offers ten different sauces of spaghetti. Grace chooses two sauces for a spaghetti. How many different possible choices could she make?
- 11 A variety pack of candy consists of nine bars each with a different flavour. If four bars of candy are chosen at random, how many different selections are possible?
- 12 In a Lotto draw, 5 different numbers are chosen from the numbers 1 to 24. How many possible selections are there? Assume that order does not matter.
- 13 A variety pack of chocolate consists of seven bars, each with a different flavour. If three bars of chocolate are chosen at random, how many different selections are possible?
- 14 Quiana is ordering a salad at a restaurant. She can order a salad consisting of just lettuce, or she can also add any of the following ingredients:
- Onions, Cucumbers, Tomatos, Cheese, Olives, Carrots, Mushrooms
- How many different variations of a salad can she possibly choose?
- 15 Out of the 16 cricket players, only 12 are selected for the team.
- a How many different teams can be made?
 - b Two opening batsmen are to be selected from among the 12 chosen players. How many combinations of opening batsmen are possible?
- 16 In court, a jury panel of 8 members is to be made up from a group of 20 candidates. In how many different ways can a panel be formed?
- 17 In a bicycle race, a quinella is a bet on the first 2 bicycles that finish the race, but the order in which these 2 bicycles place does not matter. How many different quinella bets are possible for a bicycle race where 14 bicycles are competing?
- 18 In a game of football, each team must have 1 goalkeeper and 10 other players on the field. A football squad consists of 17 players, 2 of which are goalkeepers. In how many ways can players be chosen to start the game?

- 19 A tea shop sells 10 different types of tea and 4 different sets of teacups, and has room to display 6 types of tea and 3 sets of teacups. The owner changes the display each month. How many different display combinations of tea and tea cups are possible?
- 20 A teacher wants to divide her 12 students into a group of 4, then a group of 5 and finally a group of 3. In how many ways can she do this?
- 21 Determine the number of ways can 3 cards be chosen from a standard deck of 52 cards if:
- a Exactly one of the cards is to be a Queen.
 - b None of the cards is to be a Queen.
 - c At most one of the cards is to be a Queen.
- 22 Amelia is playing a game of cards. She will randomly draw 5 cards from a standard deck of 52 cards, and wants only 3 of the cards to have clubs on them. How many different selections would contain the cards she wants?
- 23 Determine the number of ways can 6 people be chosen from a group of 11 people if:
- a The eldest person is included.
 - b The eldest person is not included.
 - c The eldest and youngest people are both to be included.
- 24 How many different unordered selections can be made from a set of 7 elements?
- 25 How many different unordered selections can be made from the set of letters of the Greek alphabet, which contains 24 letters?

- 26 In the Braille system of writing, six dots are arranged, and a particular combination of raised dots represents a particular letter. The letter 'a' is represented by one dot being raised, as shown:



- a In how many ways can two dots be raised?
- b In how many ways can 1 to 6 dots be raised?

- 27** A selection of 4 people are to be chosen from a group of 7 people. How many selections are possible if the youngest or oldest is included but not both?
- 28** A team of 3 is to be chosen at random from a group of 5 girls and 6 boys. Determine the number of ways can the team be chosen if:
- a** There are no restrictions.
 - b** There must be more boys than girls.
- 29** Frank's organisation wants to form a partnership with a charity. After a day of speaking with representatives from different charities, he looks in his wallet and sees that he has 3 identical business cards from one charity and one business card from each of the other 9 charities he spoke with. If he wants to select 3 cards to keep, no two from the same charity, how many different selections can he make?
- 30** The state government needs to determine how school funding is allocated for the following year. Of the 8 public schools that will be affected, 5 of them are asked to attend a meeting. Determine the number of ways the 5 representative schools can be chosen if:
- a** The two least funded schools must attend.
 - b** Neither of the two least funded schools can attend.
 - c** Saint Hibiscus Girls' and Saint Hibiscus Boys' are two of the public schools affected, and exactly one of them must be chosen.
- 31** 5 boys and 6 girls are part of the debate team. 4 of them must represent the school at the upcoming debate tournament. Determine the number of ways the team of 4 can be formed if:
- a** It must contain 2 boys and 2 girls.
 - b** It must contain at least 1 boy and 1 girl.
- 32** A company has recently expanded overseas and is looking to recruit 5 new Resource Analysts. They have narrowed it down to 8 equally qualified applicants, of which 2 of them can speak a second language. In how many ways can they fill the positions if exactly one of the recruits must speak a second language?
- 33** In a court case, the defendant can choose a jury of 9 members from 15 candidates consisting of 9 women and 6 men. They notice that one particular lady looks ill and wish to avoid choosing her. How many different panels of jury members can they create if they want at least 7 women to make up the jury?
- 34** A diagonal of a polygon is a line joining two non-adjacent vertices.
- a** How many diagonals does a 5-sided polygon have?
 - b** How many diagonals does a 6-sided polygon have?
 - c** How many diagonals does a n -sided polygon have?

35 Yuri constructs 13 straight lines, no two of which are parallel. Three of the lines are concurrent (intersect at a single point). How many points of intersection are there?

36 In a radio discussion about an upcoming election, 7 Labour voters, 5 Liberal voters and 5 Greens voters called up to have a say. The radio announcers want to choose 1 Labour voter, at least 4 Liberal voters and at least 3 Greens voters to talk to on air. In how many ways can they select the callers?

37 A senior science teacher wants to receive feedback from a selection of students on whether they thought the exam was fair. The table shows the range of marks and the number of students who fell within each range:

Score range	0 – 40	41 – 50	51 – 60	61 – 70	71 – 80	81 – 100
Number of students	2	4	5	7	10	5

She selects 2 students to provide feedback, but does not want more than 1 student from any score range. In how many ways can she select the 2 students?

38 10 red balls and 12 green balls are in a bag. 3 balls are to be selected from the bag at random, without replacement.

- a** Find the total number of ways that the 3 balls can be chosen.
- b** Find the total number of ways that 3 red balls can be chosen.
- c** Hence, find the probability that 3 red balls are chosen.

39 A company consists of 17 males and 12 females. 6 employees are selected at random without replacement.

- a** Find the total number of ways that the 6 employees can be selected.
- b** Find the total number of ways that 6 male employees can be selected.
- c** Hence, find the probability that 6 male employees are selected.

40 A class of 24 students has 19 people whose birthday is in April. 4 students are selected at random without replacement.

- a** Find the total number of ways that the 4 students can be selected.
- b** Find the total number of ways that 4 students can be selected such that none of their birthdays are in April.
- c** Hence, find the probability that the 4 students are selected do not have their birthdays in April.